

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T58 · Combined and Conditional Probability

Questions

Student book · questions with writing space

■ 28 pages

FAMILY MAP

Combined Conditional Probability

F01	Combine selections without replacement	9 Q 32 marks
F02	Use conditional probability and restricted sample spaces	6 Q 24 marks
F03	Combine independent events	10 Q 36 marks
F04	Combine set counts with sequential probability	1 Q 3 marks

Working Unit 14

4MA1-1H Q21

UNIT 1
3 marks

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....

(Total for Question 21 is 3 marks)

Working Unit 23

4MA1-2H Q20

UNIT 1
6 marks

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

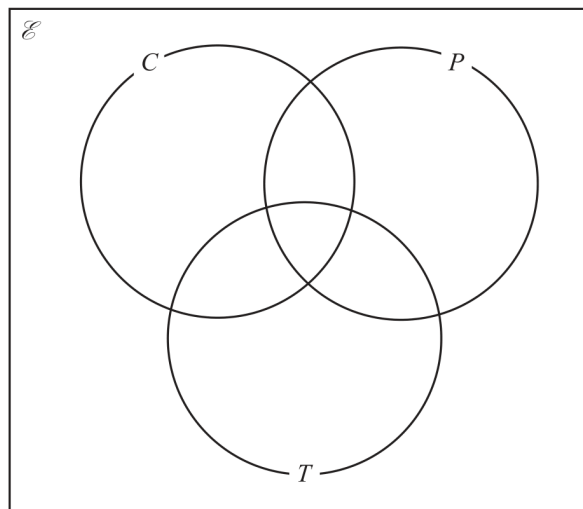
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

Question 10

4MA1-1H Q13

MODULAR UNIT 1

5 marks

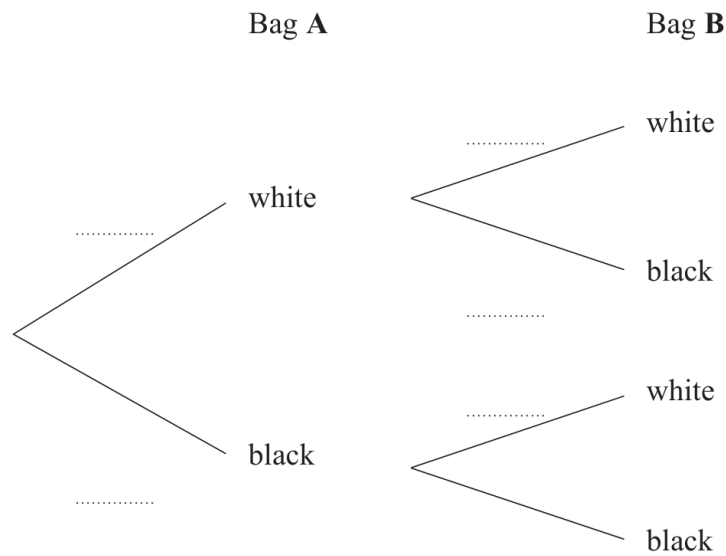
13 Harman has two bags of beads.

In bag **A**, there are 3 white beads and 7 black beads.

In bag **B**, there are 5 white beads and 4 black beads.

Harman takes at random a bead from bag **A** and a bead from bag **B**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Harman takes two beads of the same colour.

4MA1-1H Q16

4 marks

Question 14

4MA1-1H Q20 demand

MODULAR UNIT 1

4 marks

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

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Question 23

4MA1-2H Q11 (a), (b)

MODULAR UNIT 1

5 marks

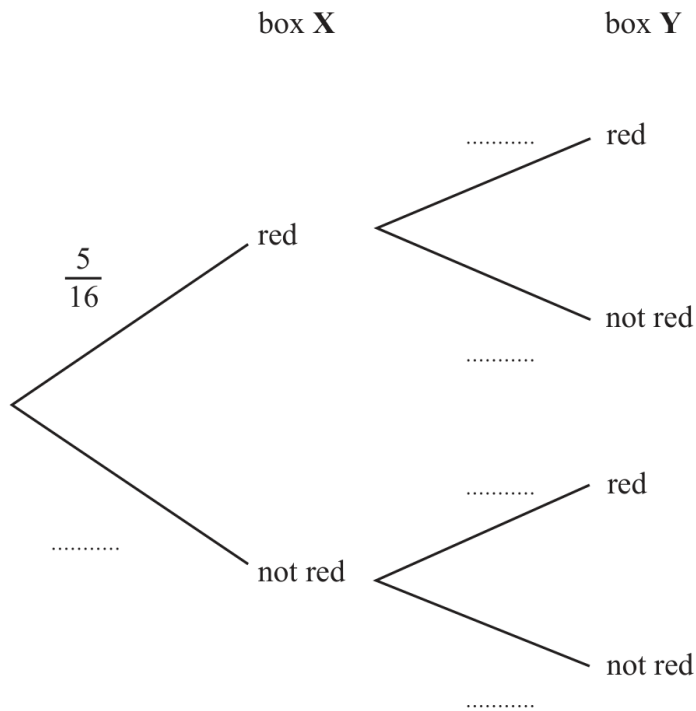
- 11** Sid has 2 boxes of crayons, box **X** and box **Y**

5 of the 16 crayons in box **X** are red.

7 of the 20 crayons in box Y are red.

Sid takes at random one crayon from box **X** and one crayon from box **Y**

- (a) Complete the probability tree diagram.



(2)

- (b) Work out the probability that Sid takes two crayons that are red or two crayons that are not red.

Question 18

4MA1-2H Q13

MODULAR UNIT 1

4 marks

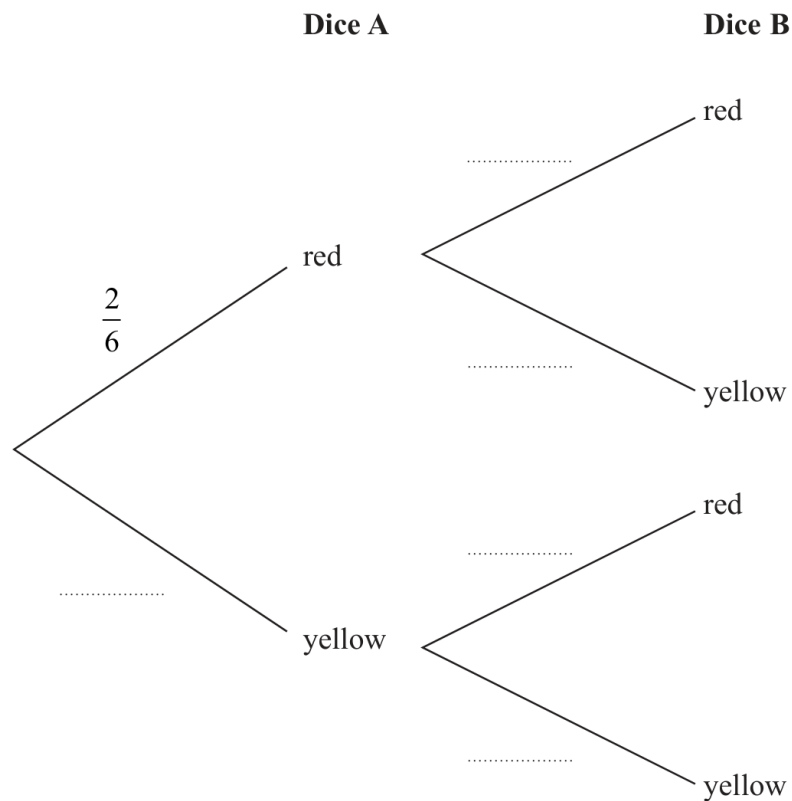
13 Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.

Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both dice land on yellow.

Question 25

4MA1-2H Q20

MODULAR UNIT 1

3 marks

20 There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

Question 09

4MA1-1H Q12 Q12(a), Q12(b)

MODULAR UNIT 1

4 marks

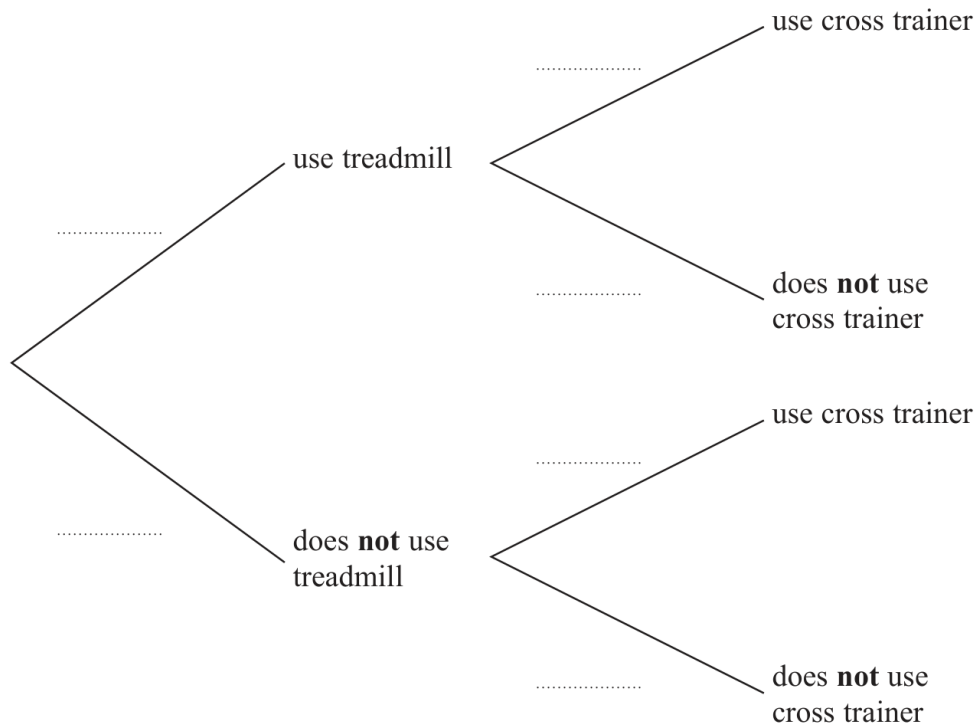
12 Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3

When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.



(2)

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

Question 10

4MA1-1H Q15 Q15

MODULAR UNIT 1

4 marks

- 15 Abraham is going to play a computer game.
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3
the probability that he draws the game is 0.5
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.

When Abraham draws a game, he scores 0 points.

When Abraham loses a game, he scores -5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

Question 24

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

MODULAR UNIT 1

8 marks

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

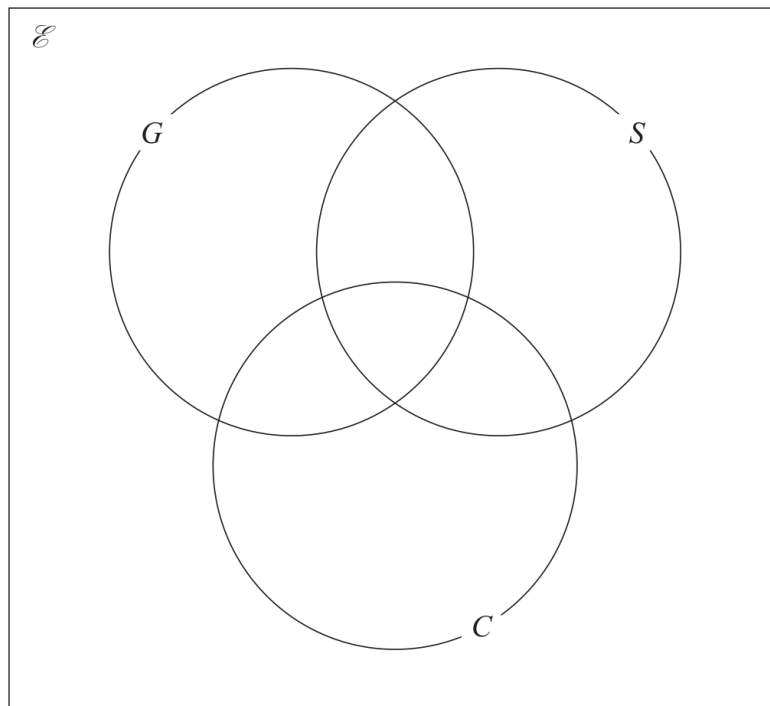
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

Question 24 (continued)

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

MODULAR UNIT 1**8 marks**

(i) $n(G)$

(1)

(ii) $n([G \cup S]')$

(1)

(iii) $n(G' \cap C)$

(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

Question 13

4MA1-1H Q14 Q14(b)

MODULAR UNIT 1

2 marks

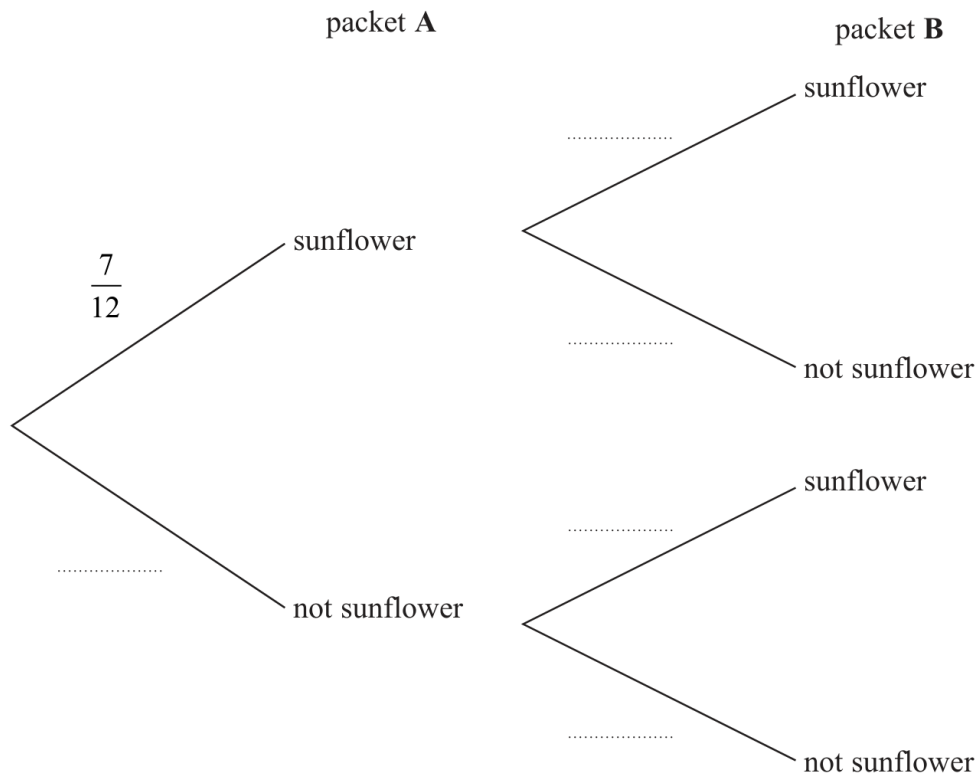
14 Aika has 2 packets of seeds, packet **A** and packet **B**

There are 12 seeds in packet A and 7 of these are sunflower seeds.

There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.

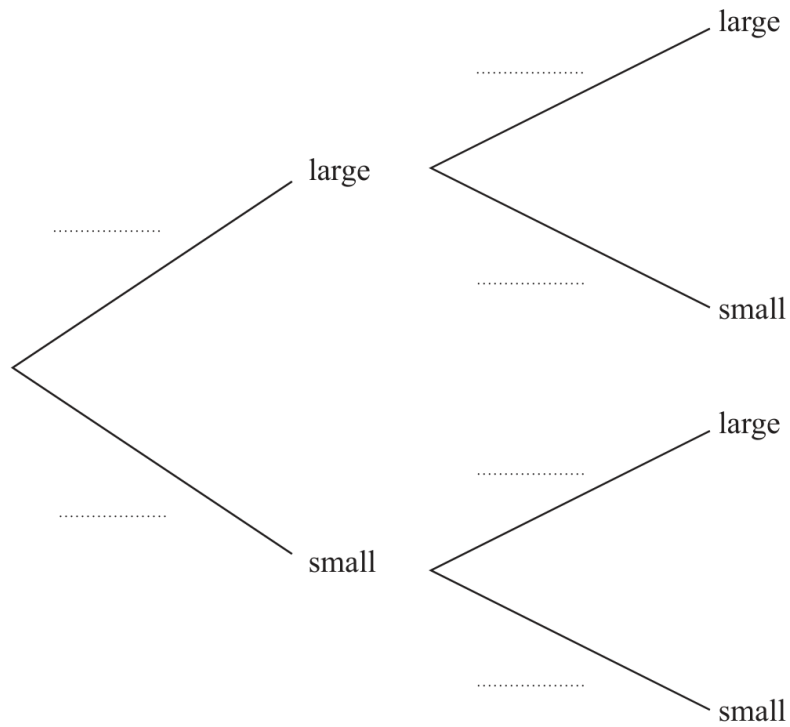


(b) Calculate the probability that Aika will take two sunflower seeds.

Work out the probability that the 5 lemon sweets are still in the bowl.

13 of the glasses are large
7 of the glasses are small

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Roberto takes two small glasses.

A histogram showing the frequency density of time taken to complete a 100m race. The x-axis is labeled 'Time (minutes)' and ranges from 0 to 50 with major grid lines every 10 units and minor grid lines every 2 units. The y-axis is labeled 'Frequency density' and ranges from 0 to 5 with major grid lines every 1 unit and minor grid lines every 0.2 units. The histogram consists of five bars, each with a width of 5 minutes. The heights of the bars are 0.7, 3.4, 1.0, 2.5, and 4.8 respectively.

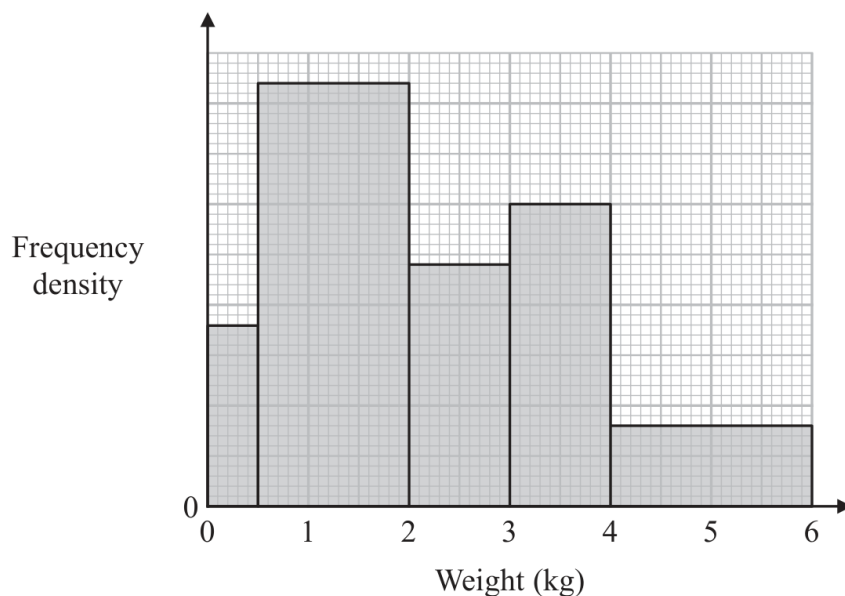
Time Interval (minutes)	Frequency Density
0 - 5	0.7
5 - 10	3.4
10 - 15	1.0
15 - 20	2.5
20 - 25	4.8

(b) find the probability that this customer spent more than 36 minutes in the supermarket.

(b) Work out the probability that Ding will win exactly one of the games.

15 A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

(3)

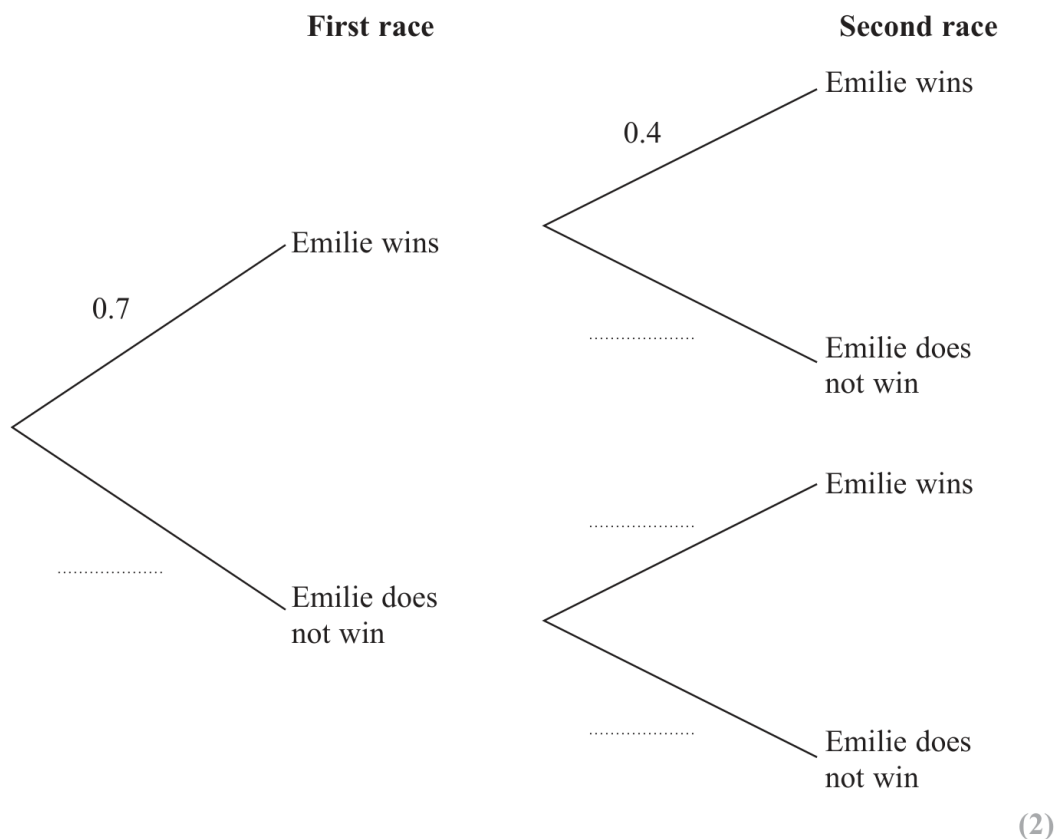
The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

13 Emilie takes part in two races.

The probability that she wins the first race is 0.7
 The probability that she wins the second race is 0.4
 The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(b) Work out the probability that Emilie wins exactly one of the two races.

(3)

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

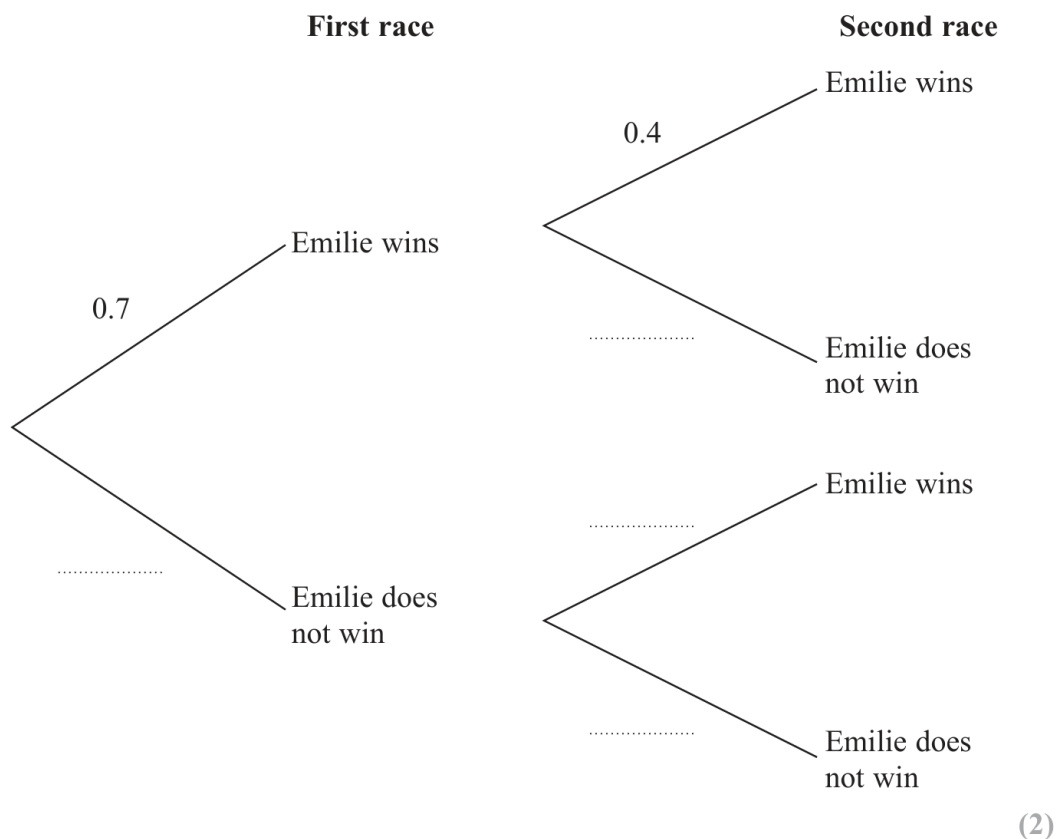
If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

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The probability that she wins the first race is 0.7
 The probability that she wins the second race is 0.4
 The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(b) Work out the probability that Emilie wins exactly one of the two races.

(3)

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If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

Show clear algebraic working.

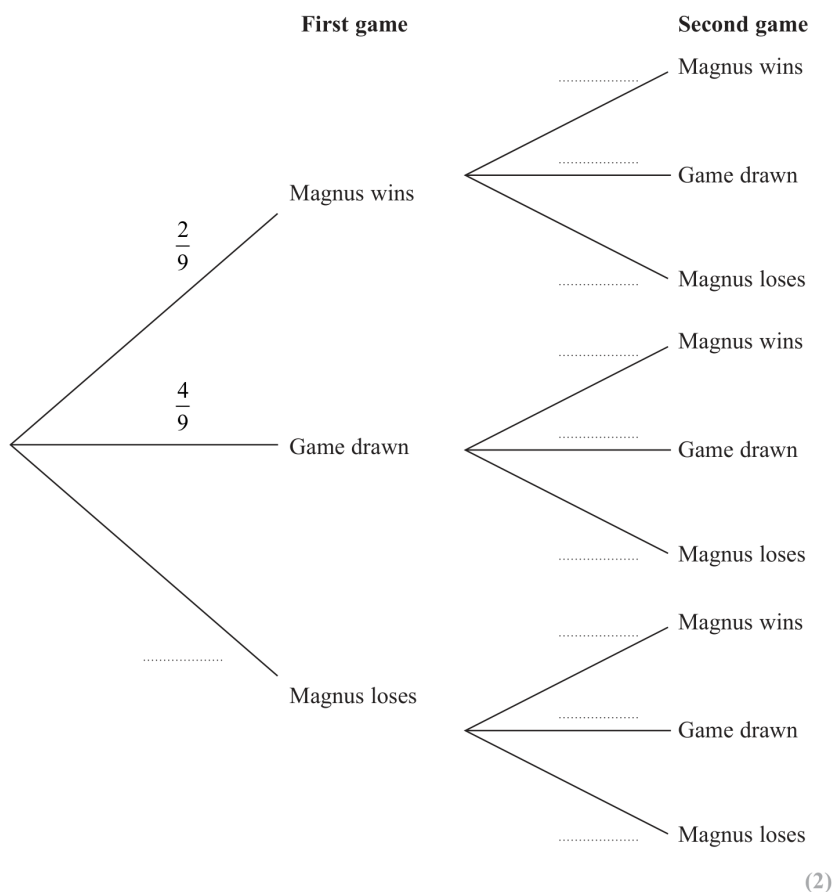
15 Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

(a) Complete the probability tree diagram.



For each game of chess,

the winner gets 2 points and the loser gets 0 points,
when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

(3)

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

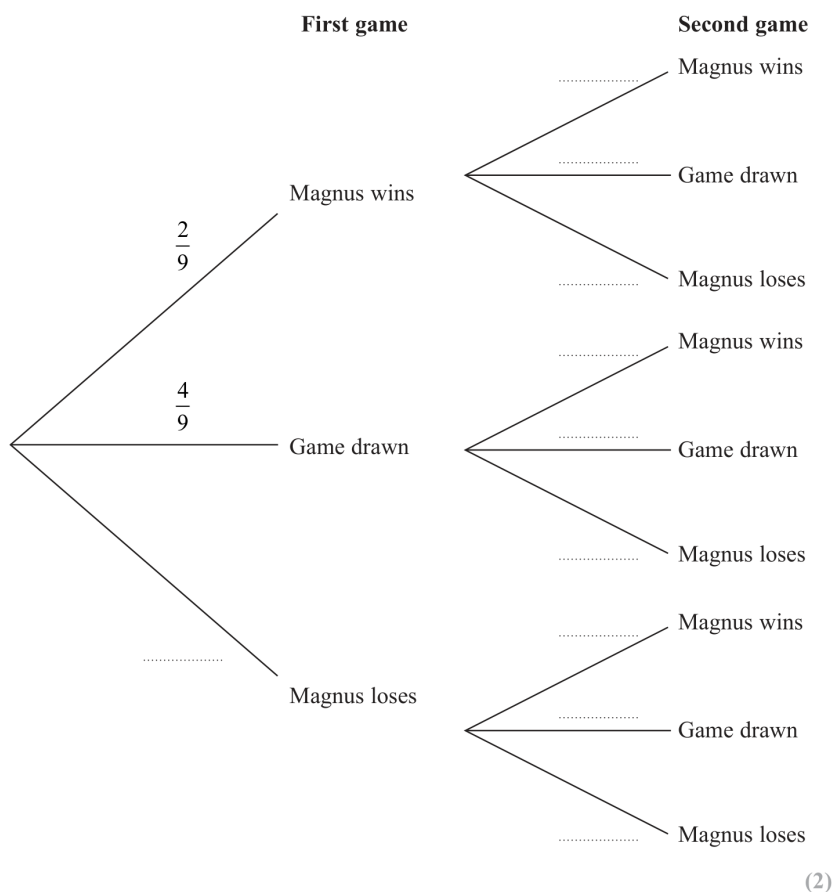
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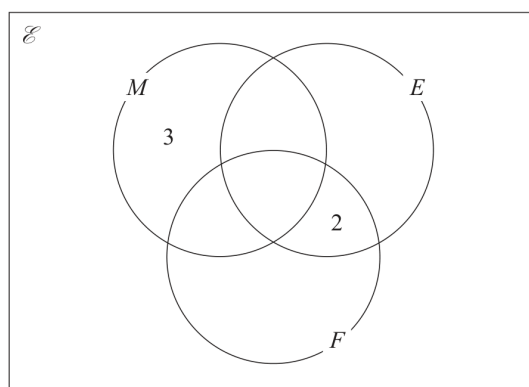
(3)

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

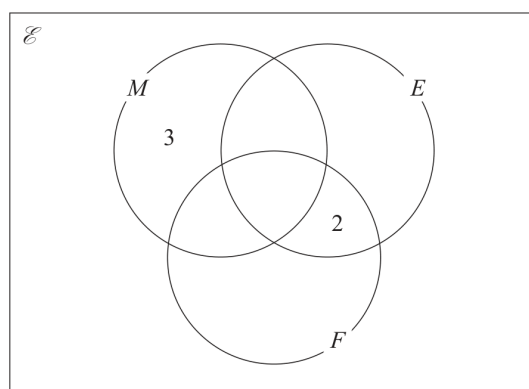
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

Chapter OL-T58: Combined & Conditional Probability

Show clear algebraic working.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Work out the probability that there is still at least one pink button in the box.